

Name:

Class:

Date:

GCSE TOPIC TEST

PHYSICS

H

Higher Tier

4.1 Energy

Materials

For this paper you must have:

- a ruler
- a scientific calculator
- the Physics Equation sheet

Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- In all calculations, show clearly how you work out your answer.

Information

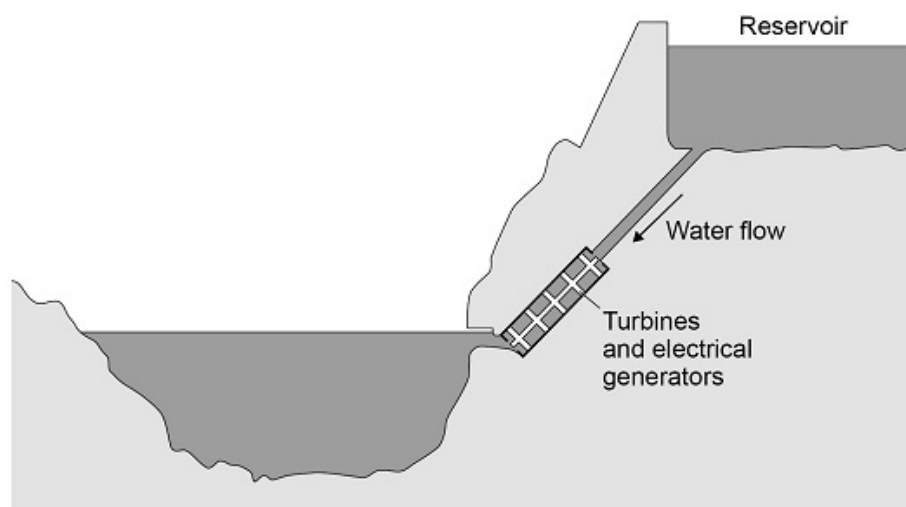
- The maximum mark for this paper is 40.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	

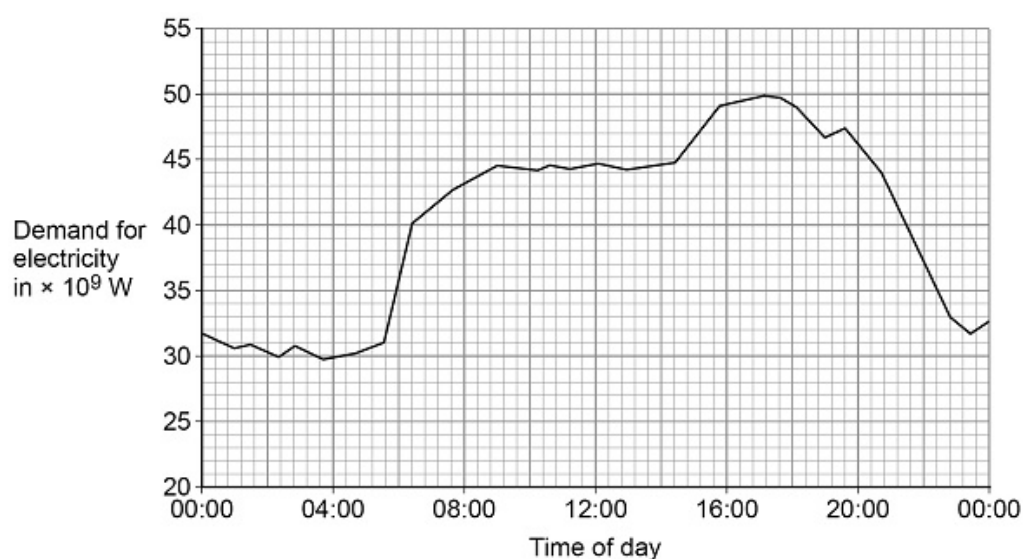
Figure 1 shows a hydroelectric power station.

Figure 1



Electricity is generated when water from the reservoir flows through the turbines.

The graph below shows how the UK demand for electricity increases and decreases during one day.



The hydroelectric power station in the above diagram can provide $1.5 \times 10^9 \text{ W}$ of power for a maximum of 5 hours.

Give **two** reasons why this hydroelectric power station is not able to meet the increase in demand shown between 04:00 and 16:00 in the graph above.

1.

2.

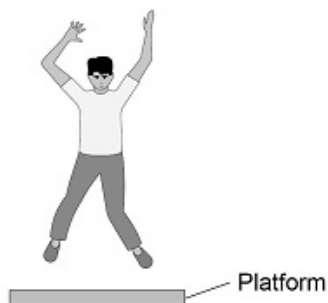
[2 marks]

A scientist investigated how the maximum muscle power of humans varies with age and gender.

The scientist asked volunteers to stand on a platform and to jump as high as they could.

Figure 2 shows a volunteer taking part in the experiment.

Figure 2



An electronic timer measured the time that the volunteer was in the air.

The muscle power in watts per kg is calculated using the following equation:

$$\text{muscle power} = \frac{9.8 \times \text{jump height}}{\text{time}}$$

One volunteer has a muscle power of 41 W/kg

He was in the air for 0.12 s

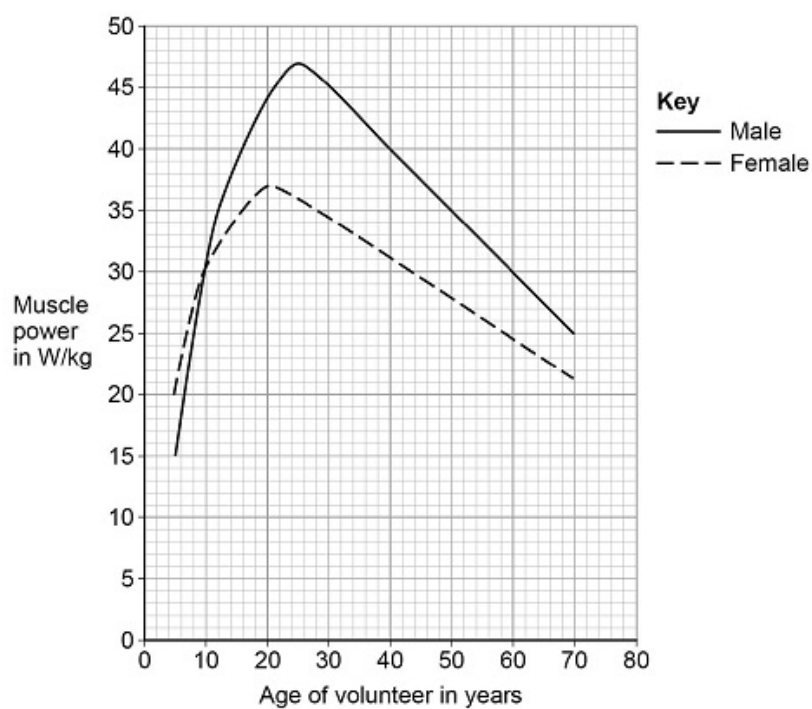
Calculate his jump height.

Jump height = _____ m

[3 marks]

Figure 3 shows the scientist's results.

Figure 3



Compare the muscle power of males with the muscle power of females.

Use data from **Figure 3** in your answer.

[4 marks]

- (a) Write down the equation which links efficiency, total power input and useful power output.

(1)

Table 1 shows information about three different light bulbs.

Table 1

Light bulb	Total power input in watts	Useful power output in watts	Efficiency
P	6.0	5.4	0.90
Q	40	2.0	0.05
R	9.0	X	0.30

- (b) Calculate the value of X in **Table 1**.

X = W

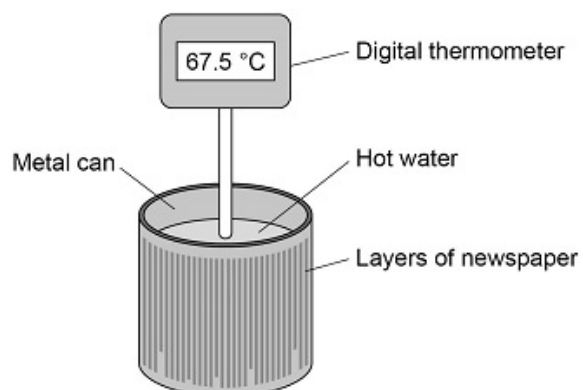
(3)

[4 marks]

A student investigated the insulating properties of newspaper.

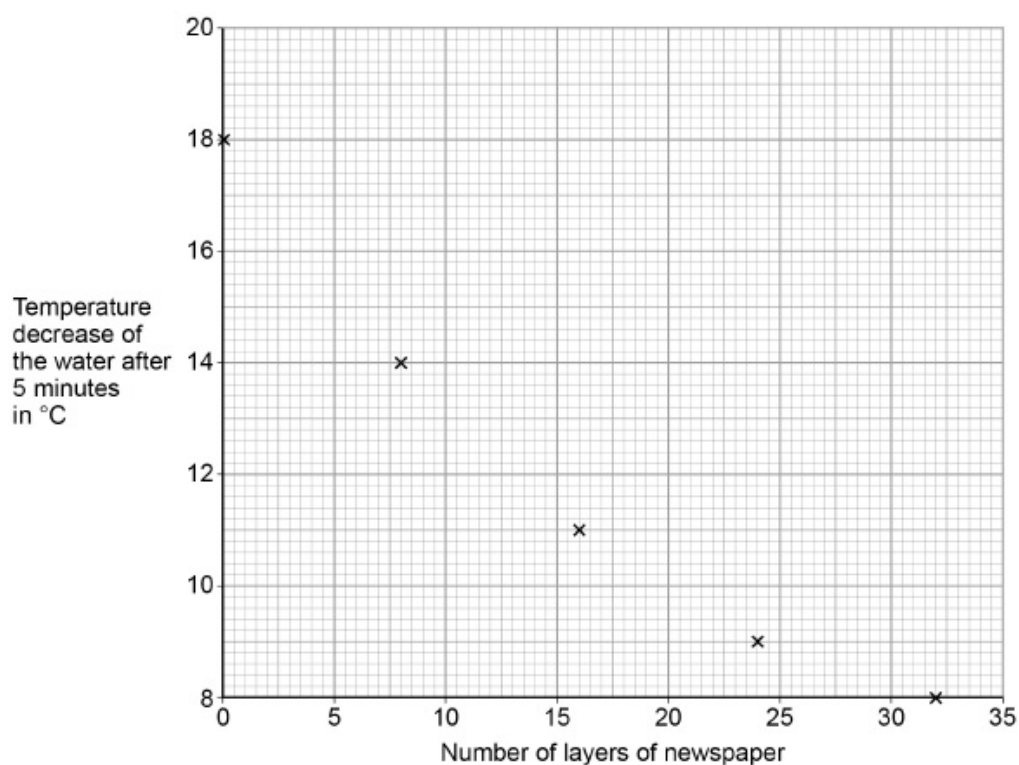
Figure 4 shows the apparatus the student used.

Figure 4



The student's results are shown in **Figure 5**.

Figure 5



Describe a method the student could have used to obtain the results shown in **Figure 5**.

04.2

Figure 6 shows the readings on the digital thermometer and the datalogger.

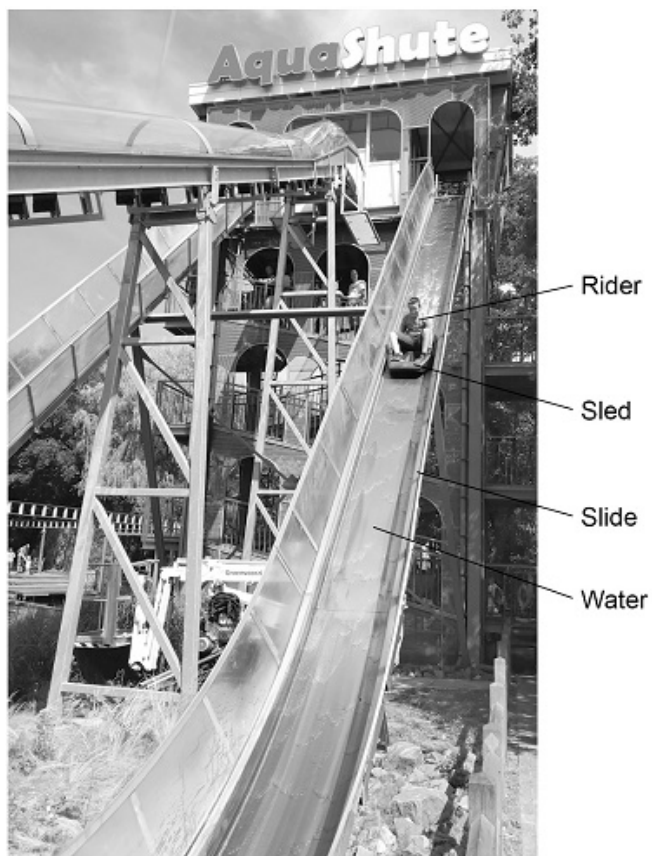


The datalogger records 10 readings every second.

Figure 7 shows a theme park ride called AquaShute.

Riders of the AquaShute sit on a sled and move down a slide.

Figure 7



A light gate and data logger can be used to determine the speed of each rider and sled.

What two measurements are needed to determine the speed of a rider and sled?

Tick **two** boxes.

- ☐ Gravitational field strength
- ☐ Length of sled
- ☐ Mass of rider and sled
- ☐ Temperature of surroundings
- ☐ Time for sled to pass light gate

[2 marks]

05.2

The decrease in gravitational potential energy of one rider on the slide was 8.33 kJ.

The rider moved through a vertical height of 17.0 m.

Gravitational field strength = 9.8 N/kg.

Calculate the mass of the rider.

Mass of rider = _____ kg

[4 marks]

05.3

At the bottom of the slide, all riders and their sleds have approximately the same speed.

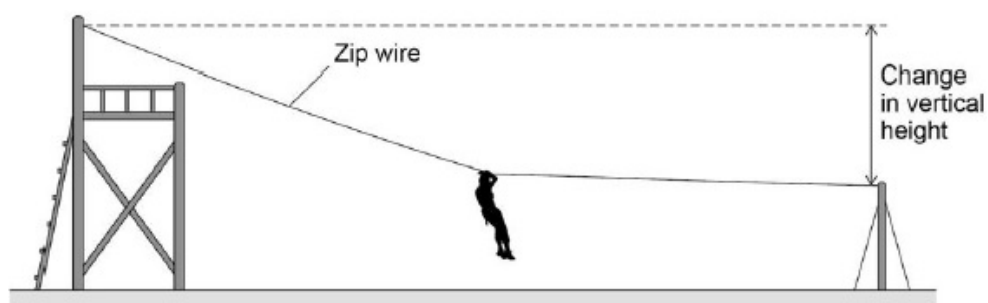
Explain why.

[4 marks]

06.1

Figure 8 shows a person sliding down a zip wire.

Figure 8



As the person moves down the zip wire, her increase in kinetic energy is less than her decrease in gravitational potential energy.

Explain why.

[2 marks]

06.2

Different people have different speeds at the end of the zip wire.

Explain why.

[2 marks]

Figure 9 shows a coffee machine. The coffee machine uses an electric element to heat water.

Figure 9



The coffee machine heats water from $20\text{ }^{\circ}\text{C}$ to $90\text{ }^{\circ}\text{C}$.

The power output of the coffee machine is 2.53 kW .

The specific heat capacity of water is $4200\text{ J/kg }^{\circ}\text{C}$.

Calculate the mass of water that the coffee machine can heat in 14 seconds.

Mass = _____ kg

[5 marks]